Report R21 — Five Encoders, and a Correction to R18

Subtype site-leakage is universal. Methylation site-leakage tracks the training corpus, not the vendor — which is not what R18 concluded.

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Every number here is substituted from `evidence/results.json` by `make_report.py`; nothing is typed by hand.

Summary

R18 measured site-leakage on two encoders and R20 on the external cohorts. Both carried the same limitation in writing: two encoders is not a survey, and one of the two was not a pathology model. Three pathology foundation models now close it — UNI, H-optimus-0 and Virchow2 — each encoding all 1,182 slides with byte-identical tile grids and its own prescribed normalisation.

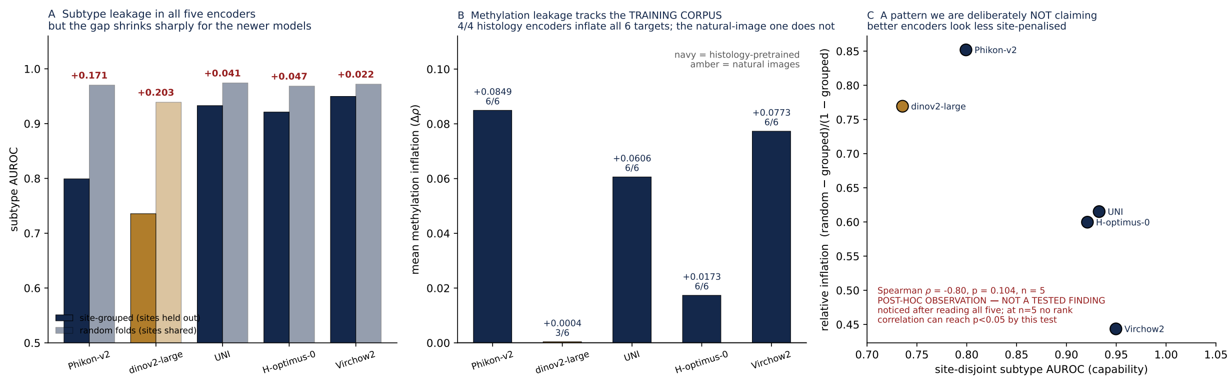
Verdict 1: SUBTYPE_LEAKAGE_UNIVERSAL. Fold assignment inflates subtype AUROC in all five encoders, every one clearing the relative-inflation bar of 0.05 that R18 declared before its own arms were read.

Verdict 2: METHYLATION_LEAKAGE_IS_A_PROPERTY_OF_HISTOLOGY_PRETRAIN
Every one of the 4 histology-pretrained encoders inflates **all six** methylation targets, with mean inflation from +0.0173 to +0.0849 in ρ. The single natural-image encoder inflates 3 of 6 with a mean of +0.0004 — indistinguishable from nothing. The dissociation is by *training corpus*, not by vendor.

This corrects R18. R18 compared Phikon-v2 against dinov2-large alone and concluded that methylation leakage was "a property of Phikon-v2 features". With four histology encoders in hand that attribution is wrong. R18 had named the limitation that produced the error — "two encoders is not a survey" — and drew the inference anyway.

1 The grid

		subtype				KEAP1	methylation		
Encoder	**Corpus**	**grouped**	**random**	**infl.**	**rel.**	**infl.**	**grouped**	**infl.**	**n**
`Phikon-v2`	histology	0.7992	0.9703	+0.1710	+0.8518	+0.0018	0.2856	+0.0849	6/6
`dinov2-large`	natural images	0.7356	0.9390	+0.2034	+0.7694	+0.0202	0.1658	+0.0004	3/6
`UNI`	histology	0.9328	0.9742	+0.0413	+0.6151	+0.0274	0.3165	+0.0606	6/6
`H-optimus-0`	histology	0.9211	0.9684	+0.0473	+0.5997	+0.0436	0.3431	+0.0173	6/6
`Virchow2`	histology	0.9497	0.9720	+0.0223	+0.4435	+0.0106	0.2914	+0.0773	6/6



A Subtype, both fold regimes. **B** Methylation inflation coloured by training corpus. **C** A post-hoc pattern, labelled as one.

1.1 Per-target methylation inflation

Target	Phikon-v2	dinov2-large	UNI	H-optimus-0	Virchow2
KEAP1 sig.	+0.0755	-0.0158	+0.0804	+0.0207	+0.0818
global	+0.0783	-0.0145	+0.0586	+0.0170	+0.0750
island	+0.1176	+0.0418	+0.0532	+0.0056	+0.0747
open sea	+0.0610	+0.0003	+0.0573	+0.0169	+0.0704
TSS200	+0.1184	+0.0048	+0.0596	+0.0294	+0.0893
gene body	+0.0589	-0.0144	+0.0545	+0.0145	+0.0727

Only `dinov2-large` has targets moving the wrong way. Every histology encoder is positive on all six, which is what makes the corpus split legible rather than a difference of degree.

2 Subtype leakage is universal but not constant

The relative figure runs from 0.4435 to 0.8518, and site-disjoint capability from 0.7356 to 0.9497. The three newer pathology models are far better at the site-disjoint task than Phikon-v2 — Virchow2 reaches 0.9497 against Phikon-v2's 0.7992 — and show correspondingly smaller inflation.

So "a fold-assignment artefact of +0.1710" is a statement about a particular encoder in 2024, not a constant of the archive. The leakage is real in every encoder tested; its size is not transferable.

3 A pattern we are deliberately not claiming

Panel C plots relative inflation against site-disjoint capability. It looks like a clean inverse relationship: Spearman $\rho = -0.80$, $p = 0.104$, $n = 5$.

We are not reporting that as a finding, and the config says so. It was noticed *after* all five encoders were read. Testing a hypothesis on the data that suggested it is the precise move this programme exists to police, and at $n = 5$ no rank correlation can reach $p < 0.05$ by this test regardless of how clean the picture looks. What it licenses is a pre-registration — fix the rule, then test it on encoders not used here — not a conclusion. It is recorded in `results.json` under `post_hoc_observation` with `STATUS` set accordingly.

4 Controls

The Phikon-v2 arms were re-run through the same harness and still match R15: subtype 0.7992 / 0.9703, KEAP1 0.6642, mean methylation inflation 0.0849. All gates in `evidence/gates.json` passed. Every arm covers the same 760 patients and 67 sites.

KEAP1 inflation stays small in all five (+0.0018 to +0.0436), which is what R15 predicted on the mechanistic grounds that mutation status has no institutional prevalence structure to exploit. A prediction made for a reason, holding across five encoders chosen for other reasons, is worth more than the same number from one.

5 Limitations

Five encoders, one archive, one cohort. TCGA lung, 760 patients, 67 sites, the same slides and the same tiles throughout.

Corpus is confounded with vintage and capacity. The three pathology models are newer, larger and trained on more data than Phikon-v2, and dinov2-large differs in corpus *and* in being the only non-pathology model. A corpus-level statement is the most parsimonious reading of Panel B, not the only possible one.

Normalisation is not held constant. H-optimus-0 prescribes its own; forcing ImageNet statistics on it would have crippled it. Each model gets its own preprocessing, which is correct practice and a departure from R18's tighter design.

CLS-only readout. Virchow2's authors recommend concatenating the class token with the mean of the patch tokens. Holding the readout constant is what makes five encoders comparable, but it may understate Virchow2's absolute capability.

Two of the five are non-commercially licensed. UNI and Virchow2 are CC-BY-NC-ND-4.0; Phikon-v2, dinov2-large and H-optimus-0 are not. Each arm records its encoder's licence in the feature files. Anything that could read as commercial should rest on the Apache-2.0 arm.

Reproducing this report

`evidence/encode_pathfm.py` (the three encoders, pinned), `evidence/mil_encoder_compare.py` (the twenty arms), `evidence/m18_five_encoder_survey.py` (this analysis), `evidence/config.yaml` (SHA-256 `b23ba9307a6081c4...`), `evidence/gates.json`, `evidence/results.json`, `evidence/arms/` (all twenty raw arm JSONs). This document: `python3 make_report.py`. Figure: `python3 make_figure.py`.